Shore blocking turns one descending sequence below an ordinal into one below its natural product

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Editorial note. The counted unit is the article's own Lemma lem:shore-blocking-wf at source lines 426-431 together with the proof environment that begins two lines after the statement ends (433-450); the pairing is the ordinary adjacent one.

Editorial note. The retained context at source line 390 is the author's own announcement of the two combinatorial lemmas and of the natural (Hessenberg) operations; the retained context at source lines 392-398 is his own definition of the natural sum and of the natural product alpha ntimes k.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, his natural-operation macros at source lines 104-107 and his quotation macro at source line 116. The LS2 font-encoding block at source lines 93-99, which declares \smashtimes through the stix fonts, is not retained because the host TeX Live has no stix directory and because the author never uses that symbol: source line 105 defines \ntimes as \dottimes and source line 110 shows the alternative already commented out. mathabx (source line 103) is retained because \dotplus, \dottimes and \dotdiv come from it. The cleveref load at source line 6 is not retained because the assembler adds its own cleveref machinery, and the inputenc load with the utf8x option at source line 8 is not retained because it fights the forced xelatex engine. Nothing was renamed, redefined or added.

Editorial note. No retained line contains a citation or a cross-reference, so the material carries no bibliography and needs no navigation pointer. In particular the sentence of the introduction that would name the proof-theoretic ordinal references is not part of the retained context.

Editorial note. The article's conservation theorems, its other combinatorial lemma lem:wf-product, its lemma lem:coh-wf-forcing-jump and lem:coh-wf-forcing-wf, and its proposition prop:coh-wf are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Natural sum and natural product of ordinals below epsilon_0. For ordinals alpha and beta below epsilon_0 written with a common family of exponents, the natural sum is obtained by adding the coefficients of like powers of omega, and alpha ntimes k is the natural sum of alpha with itself k times; the article's own definition is retained at source lines 392-398.. Fixes the two operations that the counted lemma uses in both its hypothesis and its conclusion, namely beta below alpha and the larger ordinal beta ntimes k.

- Theories RCA_0, I Sigma_1 and the well-foundedness statement WF(alpha). RCA_0 is the standard base theory of reverse mathematics; I Sigma_1 is the induction scheme for Sigma^0_1 formulas; WF(alpha) states that the ordinal alpha is well founded, and a model M satisfies WF(alpha) when it has no M-infinite descending sequence of ordinals below alpha; these are the standard notions of the reverse-mathematics literature and the article uses the notations RCA_0, I Sigma_1 and WF(alpha) throughout, for example in its abstract and introduction.. The counted lemma is stated for a model M of RCA_0 plus WF(alpha) and its proof invokes I Sigma_1 to make the constructed Turing functional total, so both theories and the meaning of the well-foundedness statement must be available.

- M[X] and the forcing extension by a real. For a model M = (M,S) of second-order arithmetic and a real X in 2^M, the article writes M[X] for the extension of M by the real X; the counted lemma is stated for every X with M[X] a model of RCA_0.. The hypothesis and the proof both quantify over reals X such that the extension M[X] satisfies RCA_0, and the proof appeals to M[X] satisfying I Sigma_1.

Before working on Π^1_1-conservation of WKL and COH, we prove two combinatorial lemma which say that Shore blocking comes for free for preserving $\mathsf{WF}(\alpha)$ for any ordinal $\alpha \leqslant \epsilon_0$. Instead of using the standard sums and products over ordinals, the natural (Hessenberg) sums and products will be more convenient:

Definition 0.1 (Natural sum and product)**.** *Let α and β be two ordinals less than ϵ_0. Let $\alpha = \omega^{\gamma_1} n_1 + \cdots + \omega^{\gamma_k} n_k$ and $\beta = \omega^{\gamma_1} m_1 + \cdots + \omega^{\gamma_k} m_k$ (We allow the n_i and m_i to be equal to 0 in order to write α and β using the same exponents γ_i). We define the natural sum of α and β (denoted by $\alpha \dot{+} \beta$) to be the following ordinal (It does not depend on the choice of the family (γ_i)) :*

$$\omega^{\gamma_1}(n_1 + m_1) + \cdots + \omega^{\gamma_k}(n_k + m_k)$$

We also define $\alpha \dot{\times} k$ to be equal to be the natural sum of α with itself k times.

Lemma 0.2. *lemma]lem:shore-blocking-wf Consider a model $\mathcal{M} = (M, S)$ and $\alpha \in M$ an ordinal less than ϵ_0. For every $k \in M$, there is a Turing functional*

Γ_k *such that, letting* $\beta \in \alpha$ *be the largest ordinal less than* α *of index at most* k*, for every* $X \in 2^M$ *such that* $\mathcal{M}[X] \models \mathsf{RCA}_0$*, if there is some* $e < k$ *such that* Φ_e^X *is an* M*-infinite decreasing sequence of elements smaller than* β*, then* Γ_k^X *is an* M*-infinite decreasing sequence of elements smaller than* $\beta \dot{\times} k$*.*

Moreover, an index of Γ_k *can be found computably in* k*.*

Proof. By twisting the Turing functionals, we can assume that for every $e, n \in M$, if $\Phi_e^\sigma(n) \downarrow$, then

(1) $n < |\sigma|$;

(2) $\Phi_e^\sigma(m) \downarrow$ for every $m < n$;

(3) $\Phi_e^\sigma(0), \Phi_e^\sigma(1), \ldots, \Phi_e^\sigma(n)$ is a strictly decreasing sequence of elements smaller than β.

Given $\sigma \in 2^{<M}$ and $e < k$, let $\alpha(\sigma, e) = \Phi_e^\sigma(s)$ for the largest $s < |\sigma|$ such that $\Phi_e^\sigma(s) \downarrow$. If there is no such s, then $\alpha(\sigma, e) = \beta$. Note that if $\sigma' \succeq \sigma$, then $\alpha(\sigma', e) \leqslant \alpha(\sigma, e)$.

Let $\sigma_{-1} = \epsilon$. Let Γ_k^X be the Turing functional which, on input n, searches for some $x > |\sigma_{n-1}|$ and some $\sigma_n \prec X$ such that $\Phi_e^{\sigma_n}(x) \downarrow$ for some $e < k$. If found, it outputs $\alpha(\sigma, 0) \dot{+} \ldots \dot{+} \alpha(\sigma, k-1)$. Note that if $\Gamma_k^X(n) \downarrow$, then by (3), $\Gamma^X(n)$ is an ordinal smaller than $\beta \dot{\times} k$.

Suppose that X is such that $\mathcal{M}[X] \models \mathsf{RCA}_0$ and there is an $e < k$ is such that Φ_e^X is total. Since $\mathcal{M}[X] \models \mathsf{I}\Sigma_1^0$, Γ^X is total.

Moreover, since $x > |\sigma_{n-1}|$, letting $e < k$ be such that $\Phi_e^{\sigma_n}(x) \downarrow$, by (1) we have $\Phi_e^{\sigma_{n-1}}(x) \uparrow$, so by (2) and (3), $\alpha(\sigma_{n+1}, e) < \alpha(\sigma_n, e)$, hence $\Gamma_k^X(n+1) < \Gamma_k^X(n)$. It follows that Γ_k^X is an M-infinite decreasing sequence of ordinals smaller than $\beta \dot{\times} k$. □